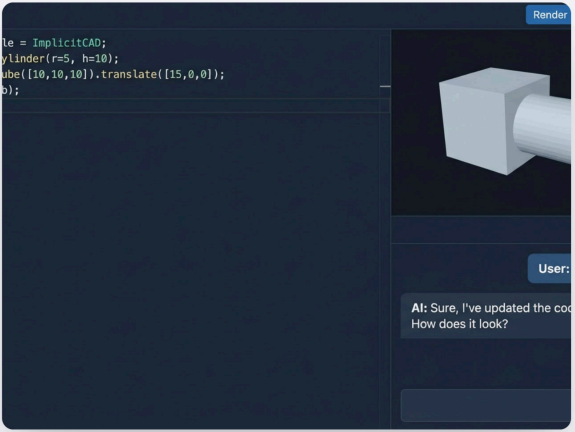
“genspark.ai” 处于全屏模式。
向下轻扫退出。

CSC392 / CSC492 PROJECT

[UofT] Print Anything

Building the Pipeline from Natural Language to 3D Object

A browser-based IDE and evaluation foundation for ImplicitCAD



PROJECT TEAM

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Why This Project?

Traditional CAD

- **GUI-heavy workflows:**
  Relies on manual mouse interaction and visual feedback.
- **Unstructured Data:**
  Complex binary formats difficult for machines to parse.
- **Hard for LLM Generation:**
  AI models struggle to output direct 3D mesh or parametric UI actions.

OUR CHOICE

Code-based CAD

- **Structured Representation:**
  Deterministic, text-based logic (e.g., ImplicitCAD, OpenSCAD).
- **Programmatic Control:**
  Enables variables, loops, and reusable modules.
- **Better for Text-to-3D:**
  LLMs excel at generating code, making this the ideal bridge for AI design.

System Overview

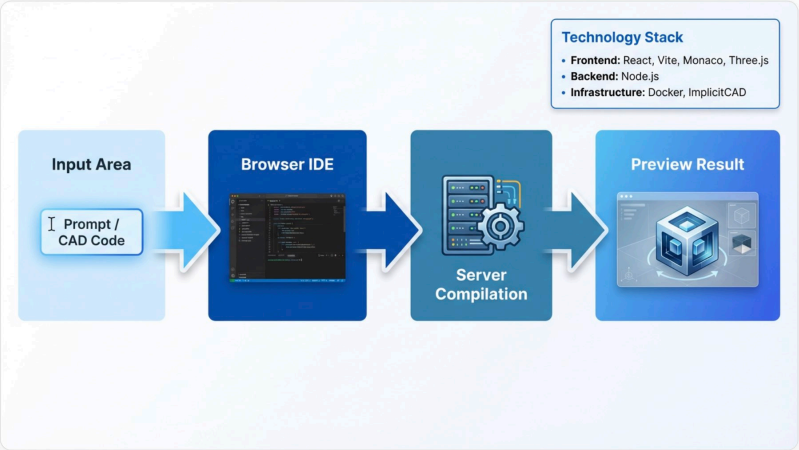


Figure 1: The complete pipeline from natural language prompt to 3D object generation

Tech Stack

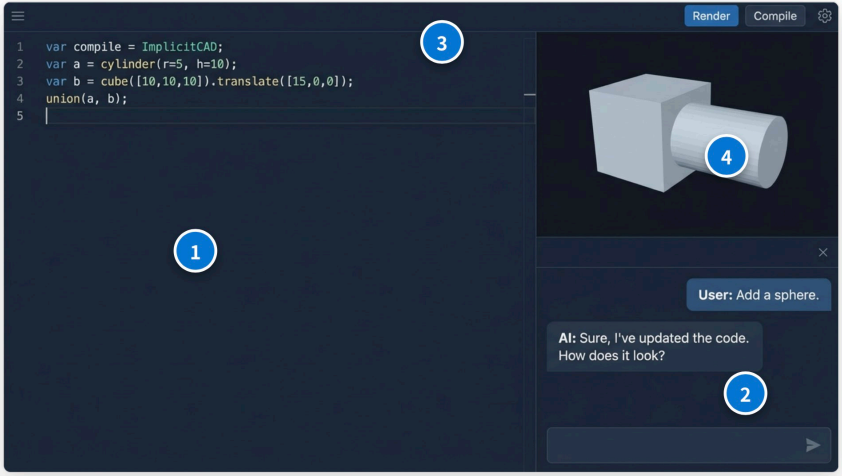
FRONTEND
- React / Vite
- Monaco Editor
- Three.js (WebGL)

BACKEND
- Node.js

INFRASTRUCTURE
- Docker
- ImplicitCAD Toolchain

EVALUATION
- SDF-based Analysis

Browser IDE



KEY FEATURES

1. **Code Editor**
   Monaco-based editor supporting ImplicitCAD syntax highlighting and validation.

2. **AI Assistant**
   Integrated chatbot for prompt-based CAD code generation.

3. **Render / Compile**
   One-click pipeline orchestration for geometry compilation.

4. **3D Viewport**
   Interactive Three.js environment for real-time model preview.

My Contribution: Test Case Design

TOTAL TEST CASES

139

Organized into 6 Major Categories

My Primary Contributions

I designed and implemented **60** specific test cases focused on complex spatial arrangements:

- **01 absolute-placement-add**
- **05 count-and-arrangement-add**

EVALUATION METHODOLOGY

Current analysis is based on **SDF (Signed Distance Function)** results to verify geometric fidelity before full mesh generation.

Detailed Breakdown by Category

Category ID & Name	Count
MY WORK **01 absolute-placement-add**	**30**
02 object-anchored-add	20
03 pattern-completion-add	24
04 bounding-box-target-add	15
MY WORK **05 count-and-arrangement-add**	**30**
06 interaction-constraint-add	20
TOTAL	**139**

Current Status, Limitations, and Future Work

Current Status

- **Browser IDE** completed with Monaco integration
- **End-to-end** workflow operational
- **139** categorized test cases implemented
- **SDF results** available for analysis

Limitations

- **ADMesh validation** is not finished yet
- Current evaluation is **not mesh-level** accurate
- SDF results alone are **insufficient** for full geometric validation

Future Work

- Add **ADMesh-based** validation to pipeline
- Build **geometry-level** comparison metrics
- Improve the **automatic evaluation** pipeline